

# Marco Gatti

Ramón y Cajal Fellow  
Institute of Space Sciences (ICE-CSIC), Barcelona, Spain  
[mgatti@ice.csic.es](mailto:mgatti@ice.csic.es) · [github.com/mgatti29](https://github.com/mgatti29) · [NASA ADS](#)

## Current Position

**Ramón y Cajal Fellow (tenure track)** January 2026–present  
Institute of Space Sciences (ICE-CSIC), Universitat Autònoma de Barcelona, Barcelona, Spain.

## Previous Positions

**KICP Fellow** July 2024–December 2025  
Kavli Institute for Cosmological Physics, University of Chicago, Chicago, USA.

**Postdoctoral Researcher** November 2020–July 2024  
Department of Physics and Astronomy, University of Pennsylvania, Philadelphia, USA. Advisors: Gary M. Bernstein and Bhuvnesh Jain.

## Education

**PhD in Physics** October 2015–September 2020  
Institut de Física d'Altes Energies (IFAE), Universitat Autònoma de Barcelona, Spain. Thesis: *Weak lensing in DES Y3: redshift distributions, shape catalogue, and mass mapping*. Advisors: Ramon Miquel and Chihway Chang. Final mark: Excellent, cum laude.

**MSc in Astronomy and Astrophysics** October 2012–July 2014  
Sapienza University of Rome, Italy. Thesis: *The growth of supermassive black holes: disk instability vs. interactions*. Advisor: Fabrizio Fiore. Final mark: 110/110 cum laude.

**BSc in Physics** October 2009–July 2012  
University of Genoa, Italy. Final mark: 110/110 cum laude.

## Selected Talks and Seminars

*Simulation-Based Inference for Weak Lensing: From Stage III to Stage IV Surveys*, DIPC, Donostia, Spain (**invited**) April 2026

*Dimensionality Reduction Techniques for Statistical Inference in Cosmology*, University College London (**invited**) December 2024

*Cosmology with weak-lensing simulation-based inference and DES*, COSMO'24, Kyoto, Japan October 2024

*Weak-lensing simulation-based inference*, Yale University cosmology seminar (**invited**) March 2024

*Non-Gaussian weak-lensing statistics in the era of precision cosmology*, Leiden University cosmology seminar (**invited**) December 2023

*Non-Gaussian weak-lensing statistics in the era of precision cosmology*, Sapienza University of Rome cosmology seminar (**invited**) December 2023

*Non-Gaussian weak-lensing statistics in the era of precision cosmology*, University of Genoa cosmology seminar (**invited**) December 2023

|                                                                                                                                                                                   |                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <i>Cosmology from weak-lensing simulation-based inference</i> , CCA workshop “Debating the potential of machine learning for astronomical surveys”, New York                      | December 2023  |
| <i>Simulation-based inference with non-Gaussian statistics of weak-lensing observables and DES Y3 data</i> , COSMO23, Madrid                                                      | September 2023 |
| <i>Constraints from non-Gaussian weak-lensing analyses</i> , Columbia University cosmology seminar ( <b>invited</b> )                                                             | March 2023     |
| <i>Cosmology with weak lensing</i> , University of California, Berkeley cosmology seminar ( <b>invited</b> )                                                                      | September 2022 |
| <i>Constraints from non-Gaussian weak-lensing analyses</i> , “Intriguing discrepancies in the growth of structures over cosmic time”, Sexten, Italy ( <b>invited</b> )            | July 2022      |
| <i>Constraints on AGN feedback models from lensing–tSZ cross-correlations</i> , SZ Workshop, Center for Computational Astrophysics ( <b>invited</b> )                             | June 2022      |
| <i>Cosmology from weak-lensing non-Gaussian statistics</i> , LMU lensing seminar ( <b>invited</b> )                                                                               | May 2022       |
| <i>Cosmology from non-Gaussian map-based statistics with DES Y3 data</i> , IberiCOS, Barcelona                                                                                    | May 2022       |
| <i>Cosmology from non-Gaussian map-based statistics with DES Y3 data</i> , YITP workshop “Cosmology with Weak Lensing: Beyond the Two-point Statistics”, Kyoto ( <b>invited</b> ) | April 2022     |
| <i>Cosmology from weak-lensing non-Gaussian statistics</i> , Lawrence Berkeley National Laboratory cosmology seminar ( <b>invited</b> )                                           | March 2022     |
| Guest expert, workshop “How measuring baryons can improve clustering analyses” ( <b>invited</b> )                                                                                 | January 2022   |
| <i>Cosmology from non-Gaussian map-based statistics with DES Y3 data</i> , Cosmopalooza 2022 ( <b>invited</b> )                                                                   | January 2022   |
| <i>Cosmology from non-Gaussian map-based statistics with DES Y3 data</i> , 239th AAS Annual Meeting ( <b>invited</b> )                                                            | January 2022   |
| <i>Clustering redshifts in DES Y3 and the DES Y3 photo-z calibration strategy</i> , GCCL seminar ( <b>invited</b> )                                                               | February 2021  |
| <i>Curved-sky weak-lensing mass map with the Dark Energy Survey Y3 data</i> , CMU/UPitt cosmology seminar ( <b>invited</b> )                                                      | September 2020 |
| <i>Curved-sky weak-lensing mass map with the Dark Energy Survey Y3 data</i> , Duke University cosmology seminar ( <b>invited</b> )                                                | January 2020   |
| <i>Cosmological constraints from moments of weak-lensing mass maps</i> , COSMO19, Aachen                                                                                          | September 2019 |
| <i>Cosmology with curved-sky weak-lensing mass maps</i> , KICP cosmology seminar, Chicago ( <b>invited</b> )                                                                      | June 2018      |
| Panelist, photo-z discussion session, “Statistical Challenges for Large-scale Structure in the Era of LSST”, Oxford ( <b>invited</b> )                                            | April 2018     |
| <i>Clustering redshifts in the DES Y1 analysis</i> , Euclid photo-z workshop, Toulouse ( <b>invited</b> )                                                                         | October 2017   |
| <i>Triggering AGN activity: disk instability vs. galaxy interactions</i> , MPE seminar, Munich ( <b>invited</b> )                                                                 | April 2015     |
| <i>The role of disk instabilities and galaxy interactions in triggering AGN activity</i> , University of Southampton seminar ( <b>invited</b> )                                   | October 2014   |

## Research Visits

|                                                                                                                |                      |
|----------------------------------------------------------------------------------------------------------------|----------------------|
| Kavli Institute for Cosmological Physics, Chicago, USA — cosmology with moments of weak-lensing mass maps      | 2018, 2 months       |
| Osservatorio Astronomico di Roma, Rome, Italy — AGN and high-energy astrophysics                               | 2013–2015, 18 months |
| University of Southampton, UK — HOD and clustering of AGN                                                      | 2014, 1 week         |
| Institute of Space Astrophysics and Planetology (IAPS), Rome, Italy — star-forming regions in Herschel surveys | 2013, 6 months       |

## Scientific Leadership, Membership, and Service

|                                                                                                      |              |
|------------------------------------------------------------------------------------------------------|--------------|
| Member, Dark Energy Survey Collaboration                                                             | 2015–present |
| Member, Euclid Collaboration                                                                         | 2022–present |
| Member, NASA OpenUniverse Collaboration                                                              | 2023–present |
| Member, South Pole Telescope Collaboration                                                           | 2025–present |
| Co-chair, Euclid Forward Modelling Work Package                                                      | 2024–present |
| Co-chair, Euclid-US monthly meetings                                                                 | 2024–2025    |
| Co-chair, DES Beyond Two-Point Working Group                                                         | 2024–present |
| Co-chair, DES Mass Mapping and Forward Modelling Working Group                                       | 2021–2024    |
| Co-chair, DES Shear Catalogue Testing Working Group                                                  | 2019–2020    |
| Mentor, DES Mentorship Programme                                                                     | 2021–present |
| External referee for MNRAS, A&A, PASJ, JCAP, ApJ, Open Journal of Astrophysics, and Nature Astronomy | 2016–present |
| External referee, UKRI Funding Service                                                               | 2024–present |
| Internal referee, Dark Energy Survey                                                                 | 2016–present |

## Awards, Fellowships, and Competitive Funding

|                                                                         |              |
|-------------------------------------------------------------------------|--------------|
| European Research Council Starting Grant, COSMO-SBI, PI, €1.5M          | 2026–2031    |
| Plan Nacional project PID2025-172756NB-C31, Co-PI, €273,850             | 2026–2029    |
| Fondo Italiano per la Scienza (FIS), PI, €1.2M (declined)               | 2025         |
| Ramón y Cajal tenure-track fellowship, €239,000                         | 2026–2031    |
| KICP Postdoctoral Fellowship, University of Chicago, \$246,000          | 2024–2027    |
| Postdoctoral Fellowship, University of Pennsylvania, \$310,000          | 2020–2025    |
| GCCL Postdoctoral Fellowship, University of Bochum, €220,000 (declined) | 2020         |
| Italian habilitation to academic professorship                          | 2023         |
| Dark Energy Survey Builder status                                       | 2021–present |
| Severo Ochoa “la Caixa” PhD Fellowship, €100,000                        | 2015–2019    |
| “Wanted the Best” MSc Fellowship, Sapienza University, €5,000           | 2013         |
| “Percorso di Eccellenza” MSc Fellowship, Sapienza University, €2,000    | 2013         |

## Computing Allocations

|                                                                                   |              |
|-----------------------------------------------------------------------------------|--------------|
| PI, Euclid-US NERSC allocation: 40,000 GPU hours and 5 million CPU hours per year | 2025–present |
| PI, Argonne Leadership Computing Facility: 150,000 GPU hours                      | 2024–present |
| Co-PI, DiRAC: 140,000 GPU hours                                                   | 2024–2026    |

## Supervision and Mentoring

|                                    |              |
|------------------------------------|--------------|
| Simone Vinciguerra, PhD researcher | 2025–present |
|------------------------------------|--------------|

|                                                                 |              |
|-----------------------------------------------------------------|--------------|
| Kyle Boone, PhD student, Harvard University                     | 2024–present |
| Alan Zhou, PhD student, University of Chicago                   | 2024–present |
| Johnny Pitocco, PhD student, University of Chicago              | 2024–present |
| Kunhao Zhong, PhD student, University of Pennsylvania           | 2023–2024    |
| Dhayaa Anbajagane, PhD student, University of Chicago           | 2023–2024    |
| Shub Agrawal, PhD student, University of Pennsylvania           | 2023–2024    |
| Minsu Park, PhD student, University of Pennsylvania             | 2022–present |
| Vernon Wetzell, PhD student, University of Pennsylvania         | 2022–2025    |
| Rafael Gomes, PhD student, University of Pennsylvania           | 2022–2025    |
| Conghao Zhou, PhD student, University of California, Santa Cruz | 2022–2025    |
| Shivam Pandey, PhD student, University of Pennsylvania          | 2020–2022    |
| Giulia Giannini, PhD student, Universitat Autònoma de Barcelona | 2020–2022    |
| Four undergraduate students, University of Pennsylvania         | 2020–2024    |

## Outreach

---

|                                                                                                                          |                           |
|--------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Astronomy Conversations Presenter, Adler Planetarium, Chicago — monthly public discussions about astronomy and cosmology | 2024–2025                 |
| Press release and media interviews for the DES Year 3 analysis                                                           | May and October 2021      |
| Hands-on cosmology activities for high-school students visiting IFAE                                                     | March 2018 and March 2019 |
| Cosmology seminar for high-school students visiting IFAE                                                                 | April 2017                |